

Haskell in the Newsroom

A little background

- Twitter: @erikhinton
- Been at the Times for 2.5 years
- Been using Haskell almost as long as I've been programming

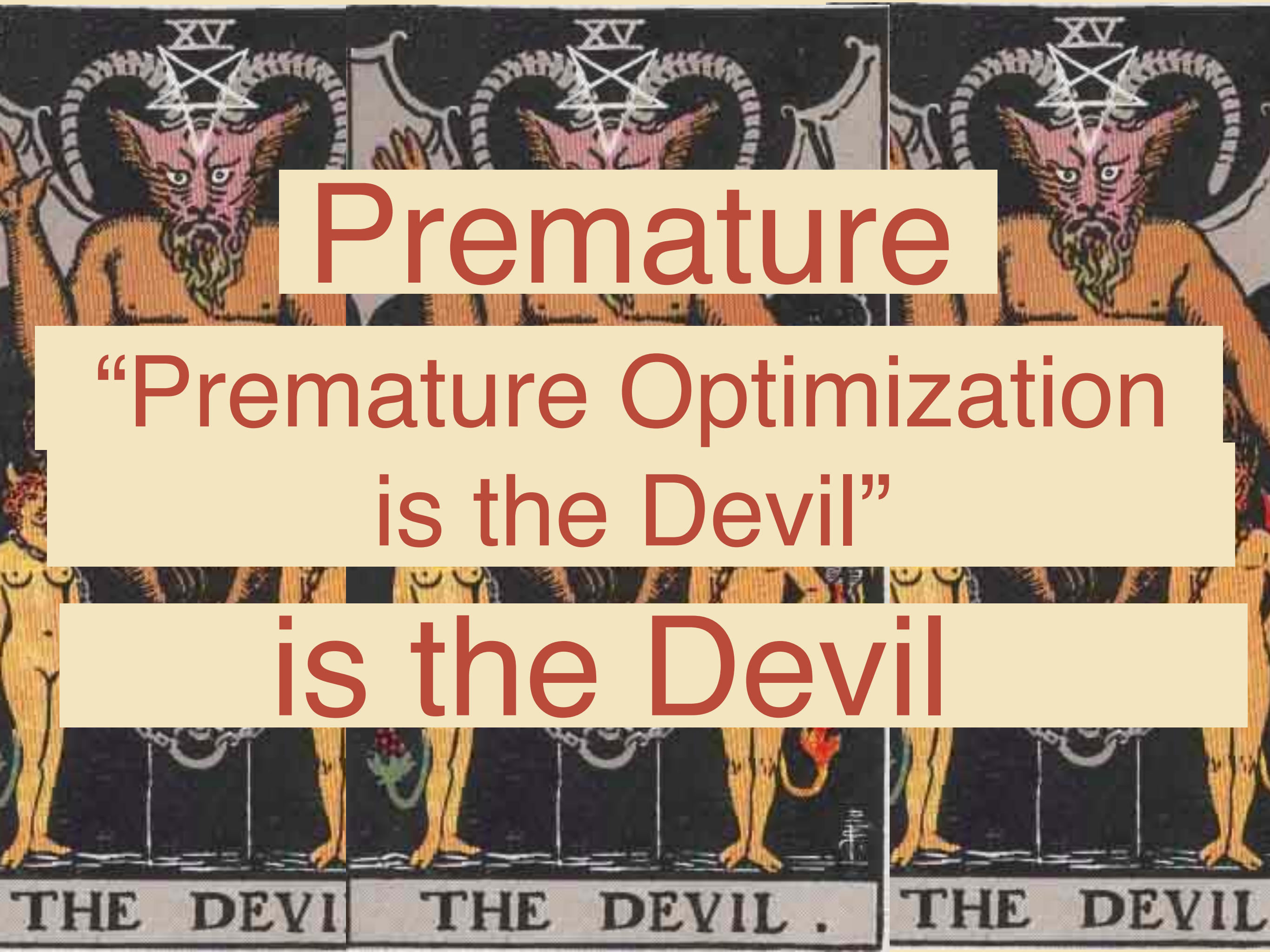


What to expect

- A perspective on integrating Haskell into a newsroom
- A paucity of actual code (not the point)
- The first google image I could find for each slide



**“Premature Optimization
is the Devil”**

The background of the image consists of three identical tarot cards, specifically 'The Devil' card from the Rider-Waite-Smith deck. Each card features a red devil with horns and a pentagram on its forehead, standing on a cross. The cards are arranged side-by-side, with the central card slightly offset. The text is overlaid on these cards.

Premature

“Premature Optimization is the Devil”

is the Devil

THE DEVI

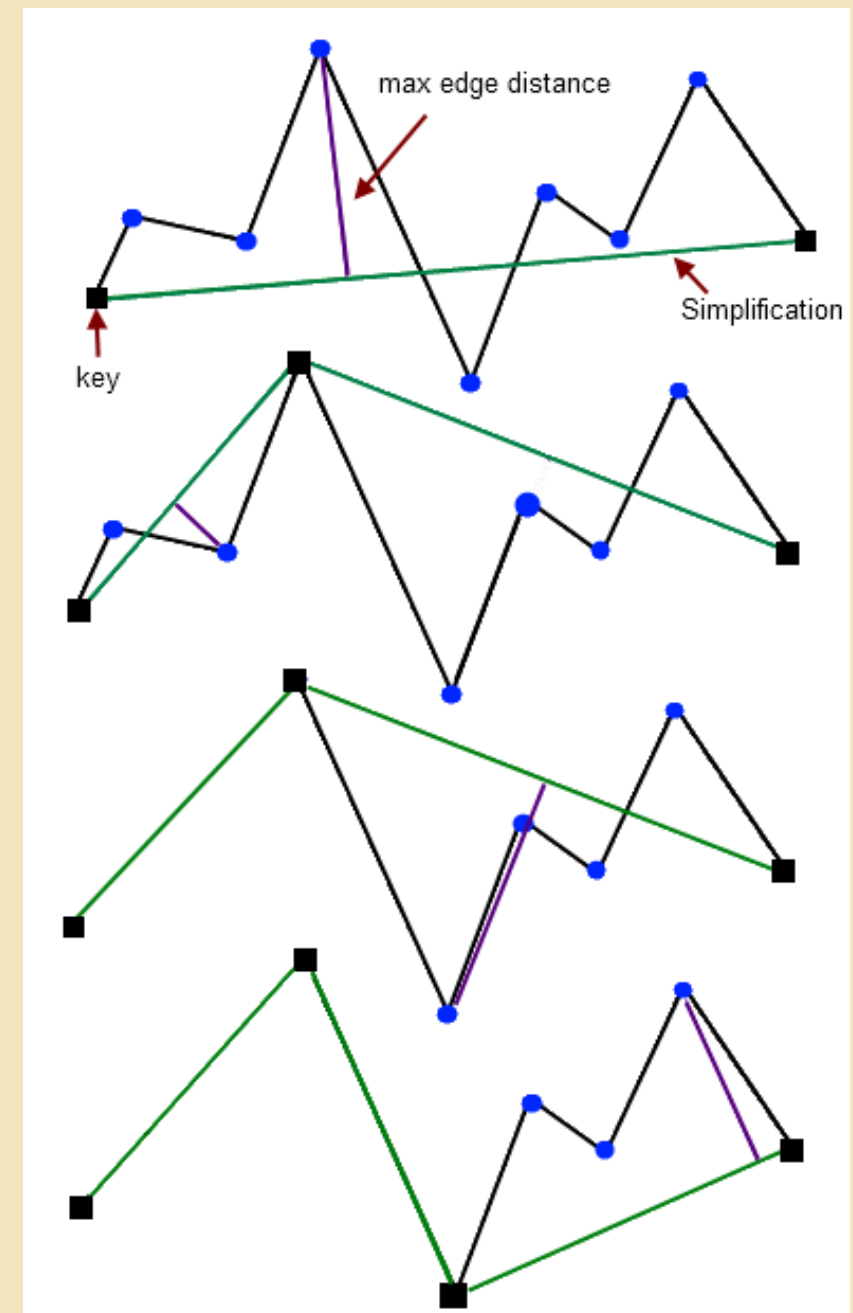
THE DEVIL .

THE DEVIL

How it all began

P e u c k e r m y b r a i n s o u t

- “This algorithm seems simple, should be easy to whip up in Ruby.”
- ERRNO - How about I just crash on any moderately large shapes
- “I should learn Haskell.”



The agony of insulation

- You have dig into the Ruby source C (MRI) to even find the time complexity of array operations
- Insulated from the cost of our code, we lowball and undermine our ambition (in



What Haskell offers

`E x t r a !   E x t r a !`

- Raw speed
- Data analysis parallelism
- Simple executables (No VM)
- Constant-space, resource-deterministic, modular stream processing
- Data abstraction

Speed

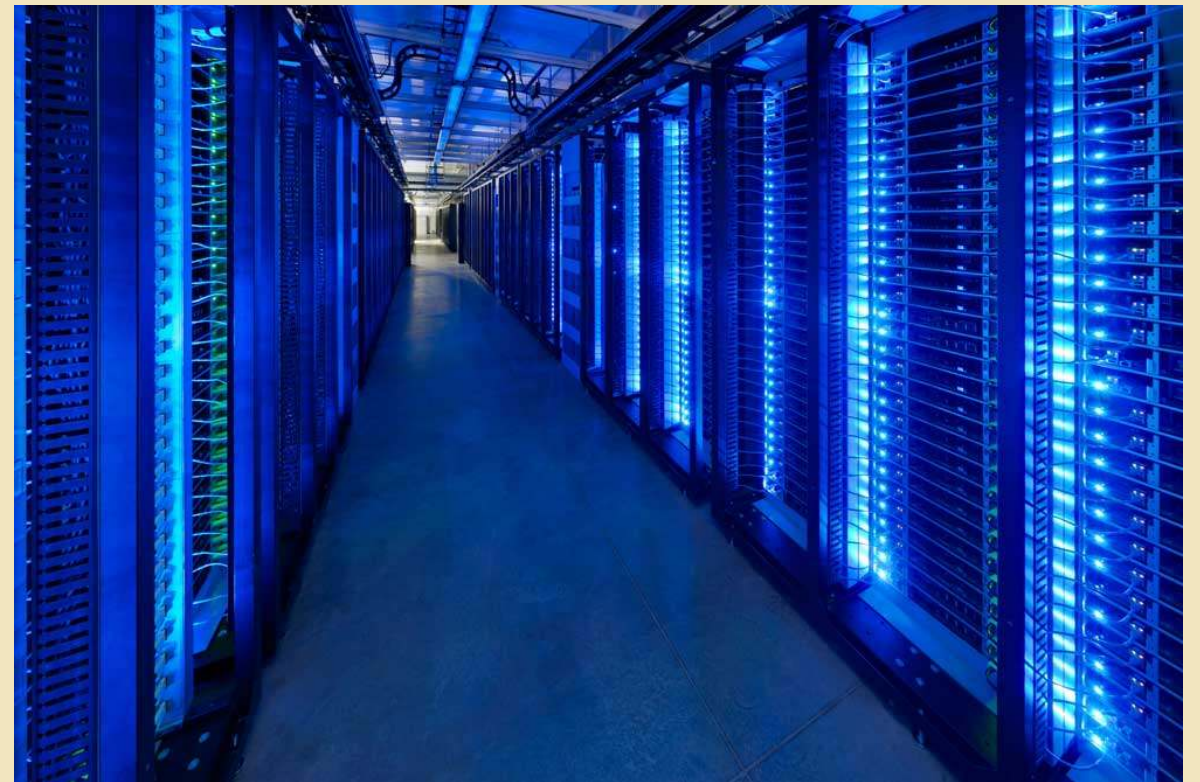
...and its discontents

- Not really a claim about benchmark speed (though true)
- Largely a product of hyper-focused libraries/structures
- Naive code on deadline - 60+ apps/yr
- Ruby's insanely expensive ORMs



Throw EC2 at it?

- There's a misconception that since we can boot up XXL compute clusters, offline processing doesn't need to be optimized
- The cost of dev time
- Newspapers are poor(ish)
- Creative iteration



But ... queues!

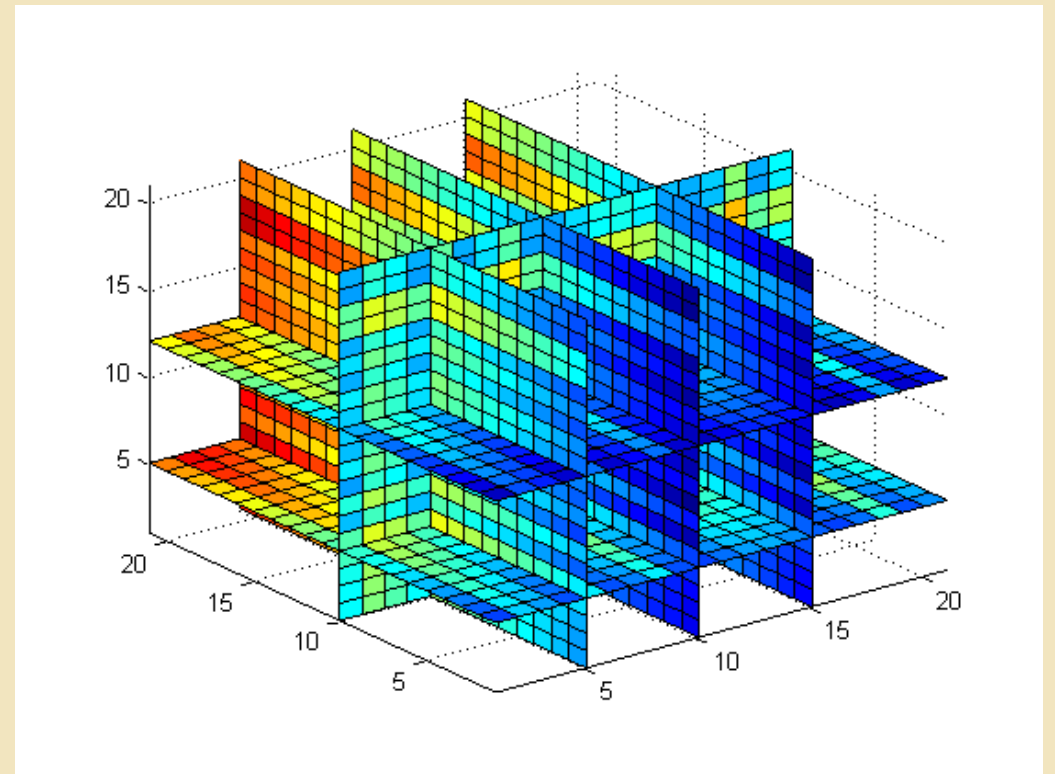
- Now you have two problems
- Distributed systems are never as simple as they seem
 - See Election Night



Parallel Dimensions

A R e p a l o v e s t o r y

- Single-machine
- Automatically parallel operations with regular, multi-dimensional arrays
- Moves data journalism out of get-a-coffee wait times

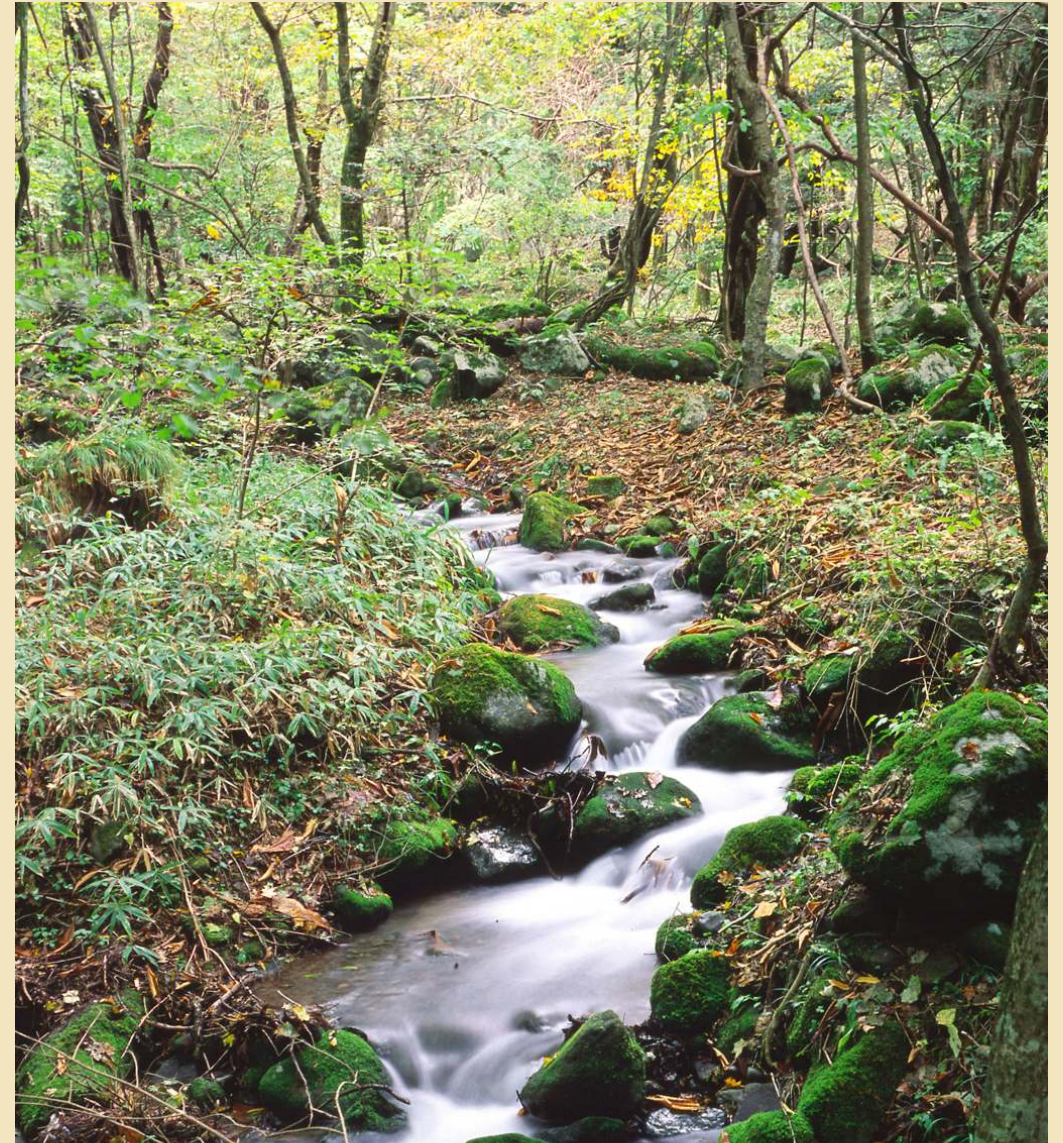


Flirting with a lisp

- Clojure, Java, and Scala are other languages we have used or tried to use for these data processing tasks.
 - Riemann
- The “complexities” of JVM-ed scripts on a cron/on a worker
- Daemons for short-lived apps
- Files over folders

Streaming data

- The mythical 400-gigabyte XML file
- Olympics, election
- Everyone loves a good firehose



Why conduits

- Both Ruby and Python offer constant-space file stream consumption with iterators that perform nicely
- Modularity
- Moving from fixture to real data, fast prototyping
- What about web streaming?
- “Use EventMachine/ Twisted”
- Handle low-level details with ‘socket’ or ‘httplib’
- In conduit: just replace source with http-conduit

Can-do-its

```
EventMachine.run do
```

```
  http = ...
```

```
  http.stream do |chunk|
```

```
    File.open("foo.txt","a"){|f|
```

```
      f.write(chunk)}
```

```
    puts chunk
```

```
end
```

```
sourceList ["1","2","3"] $$
```

```
  conduitFile "foo.txt" =$
```

```
  sinkHandle stdout
```

```
responseBody res $$+-
```

```
  conduitFile "foo.txt" =$
```

```
  sinkHandle stdout
```

Can-do-its Pt. 2

```
conduit :: Conduit Int IO String
```

```
conduit = do
```

```
-- Get all of the adjacent pairs from the stream
```

```
mi1 <- await
```

```
mi2 <- await
```

```
case (mi1, mi2) of
```

```
  (Just i1, Just i2) -> do
```

```
    yield $ show (i1, i2)
```

```
    leftover i2
```

```
  conduit
```

[https://www.fpcomplete.com/school/advanced-haskell/
conduit-overview](https://www.fpcomplete.com/school/advanced-haskell/conduit-overview)

Data, abstraction

- The purists' comma
- “We should stop focusing on writing functions to pull specific data out of objects, and start leaning on the structure of our data.”
- The olympics data horrorshow



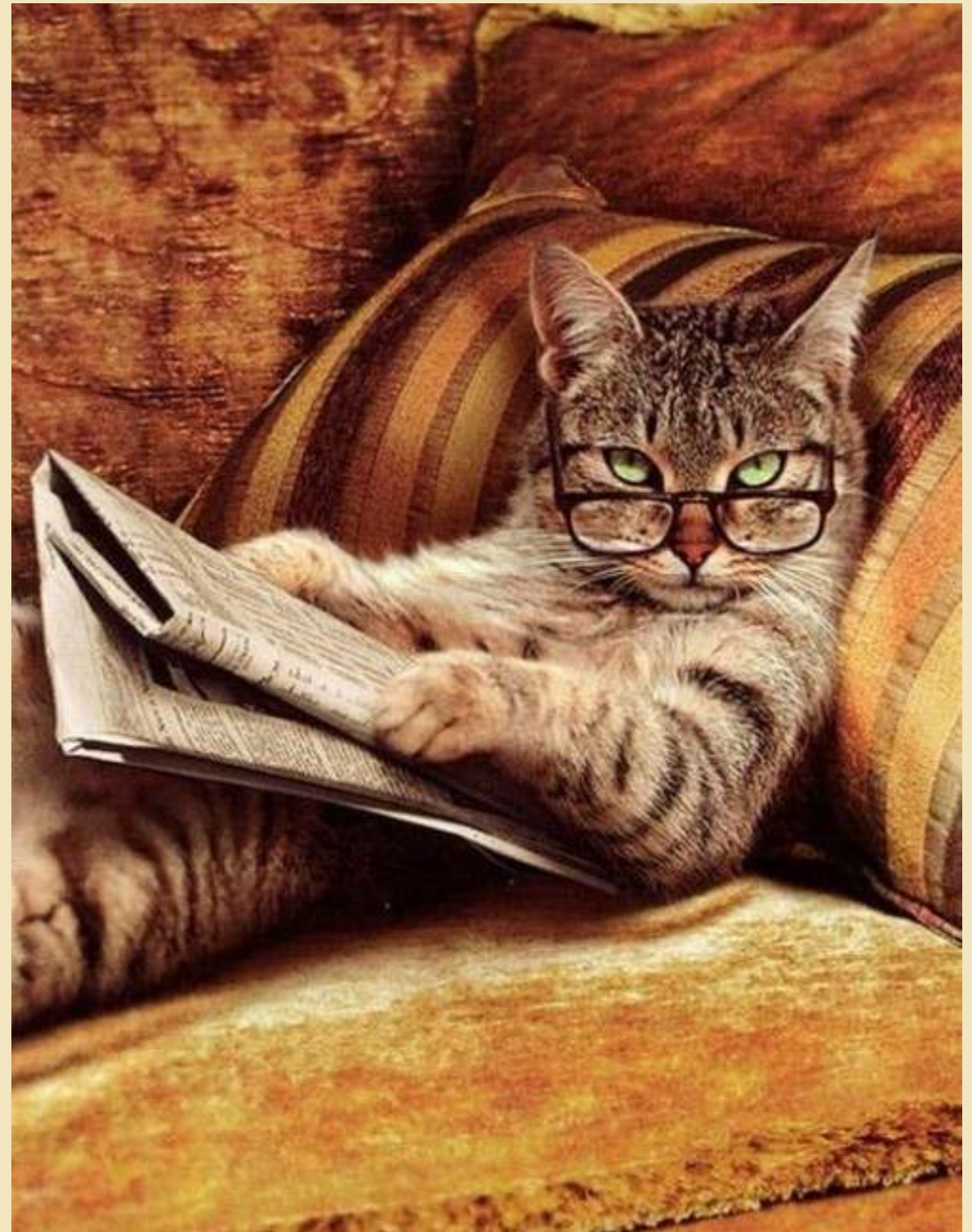
Olympic pains

- A bag full of helper functions, ad hoc parsers
- Parsing/flattening data vs. querying a known structure
- Adventures in non-composability



Lenses

- Specify and spell out the abstract structure/relationships in our data
- Free reference!
- Looping over objects and getting/setting sub-values vs. functional combinators



Lenses

A l i s t o f r e c o r d s w i t h n u l l s

```
things.each do |t|
  if t
    t[:results].each do |r|
      r[:done] = true
    end
  end
end
```

```
things.map do |t|
  if t
    t[:results].map do |r|
      r[:done]
    end
  end
end
```

```
{- assume datatypes are above -}
done_t :: Traversal' [Maybe Record] Bool
done_t =
  traverse._Just.results.traverse.done

set (done_t) True things
toListOf (done_t) things
```


Our first foray

All the tweets

- Previously: a script running on a cron (the newsroom standard)
- But what of hurricanes!?
- Sunday routines
- Could n-gram at speed of firehose



A Day of Words

“I’ll take it”

- Lick & Twobble
- Lick: enumerator, hashmap
- Twobble: enumerator, attoparsec, and aeson, never used
- Low bars and first runs: “We already use Haskell”



Fashion Fingerprints

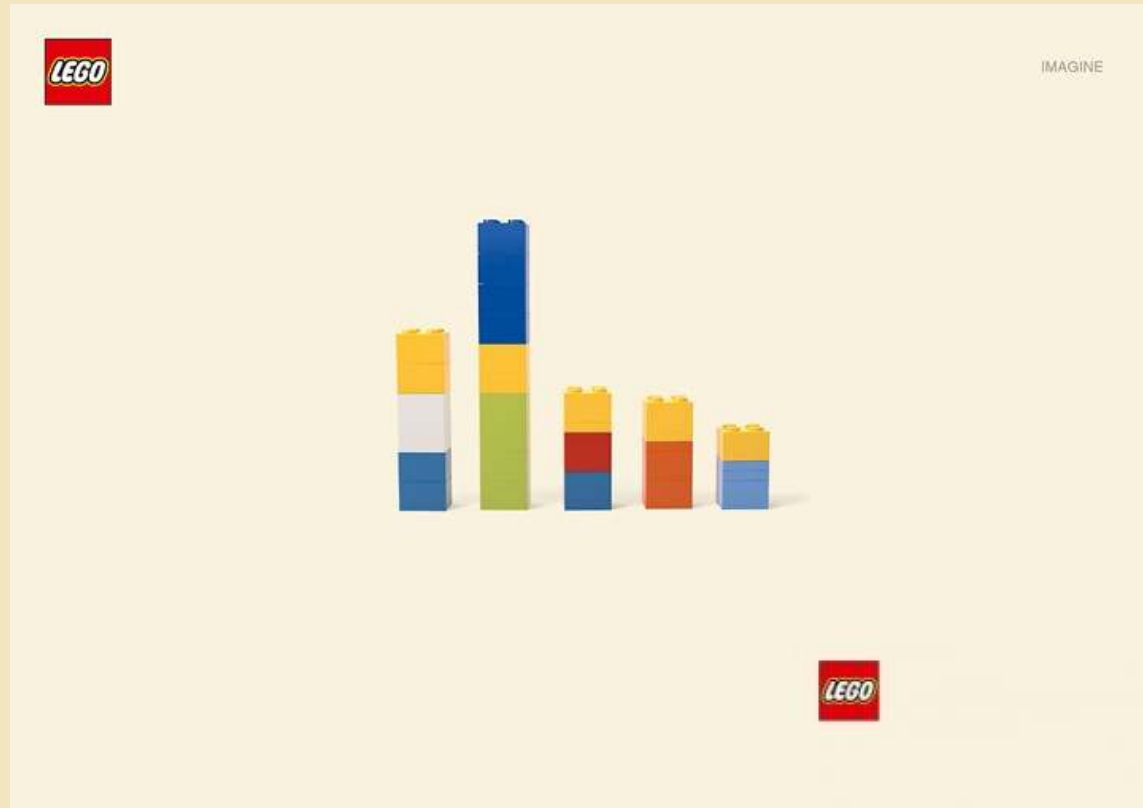
The problem

- Fashion week handwaving
- “I wish you’d do more Nate Silver stuff”
- No one understands fashion week



The Inspiration

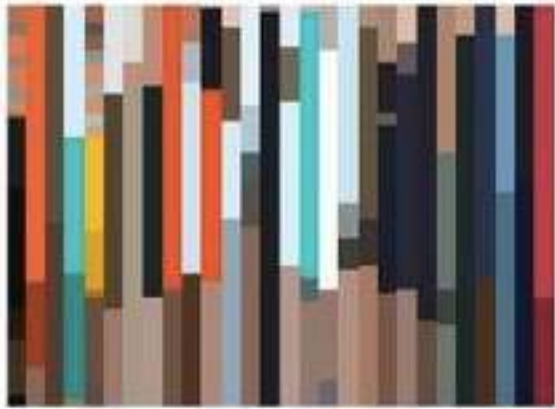
J u n g v o n M a t t



Fingerprints



Marissa Webb



Costello Tagliapietra



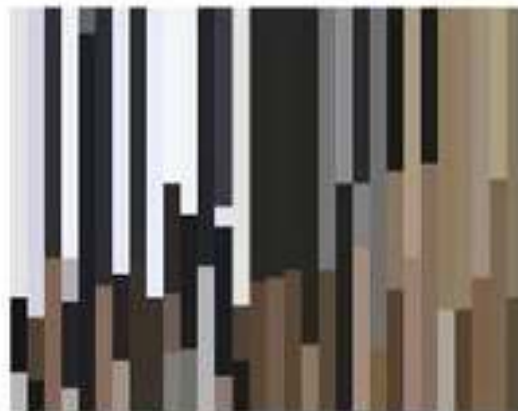
Tanya Taylor



Creatures of the Wind



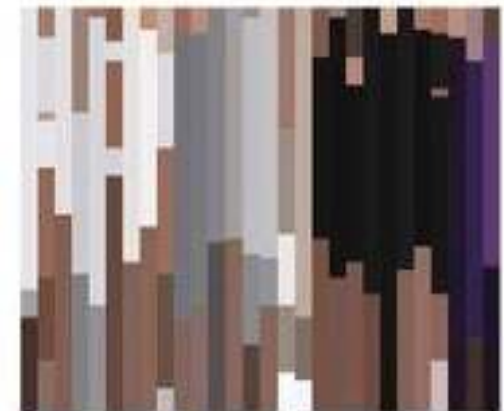
Jason Wu



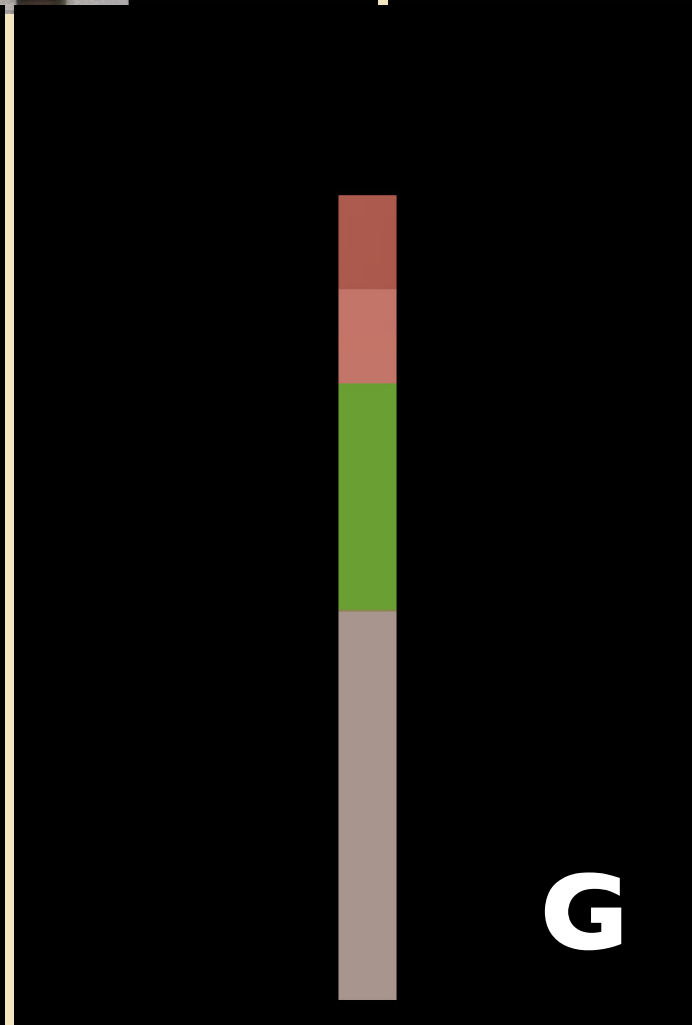
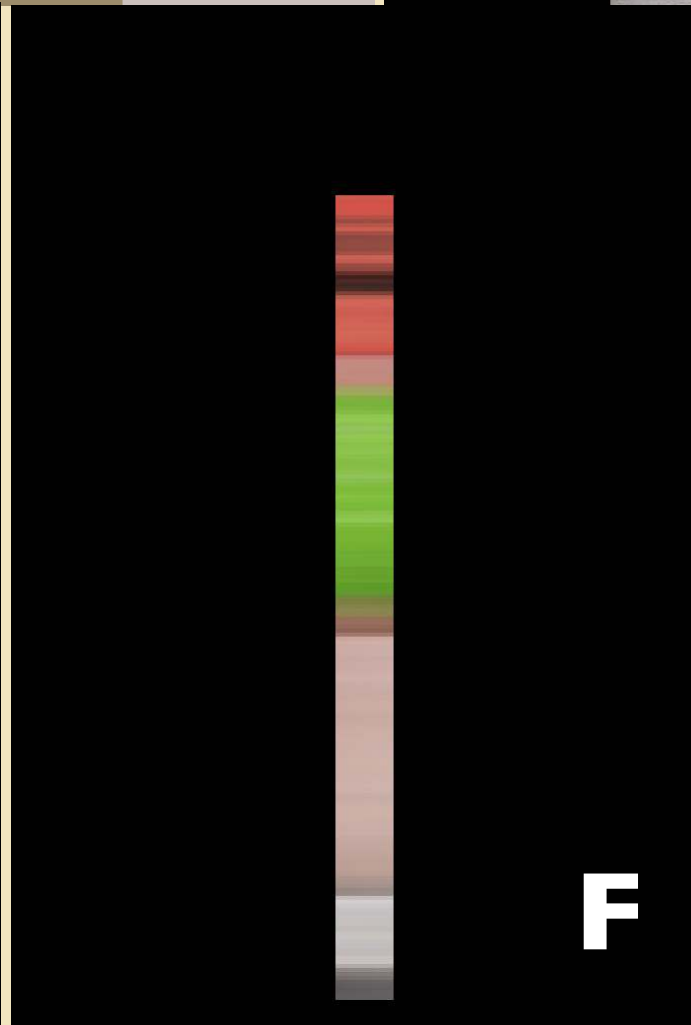
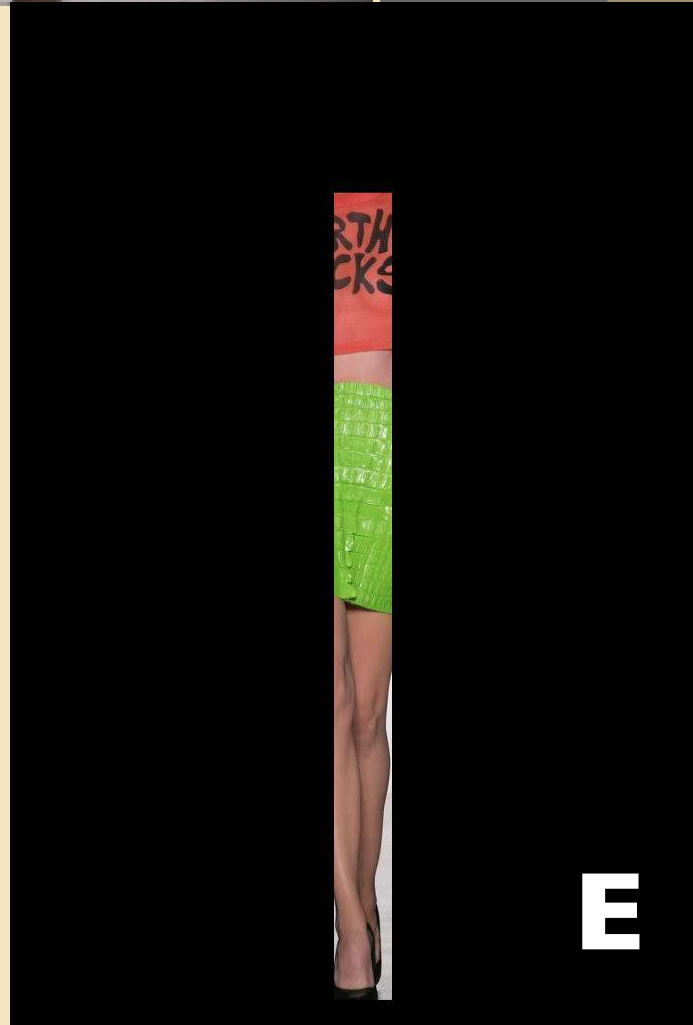
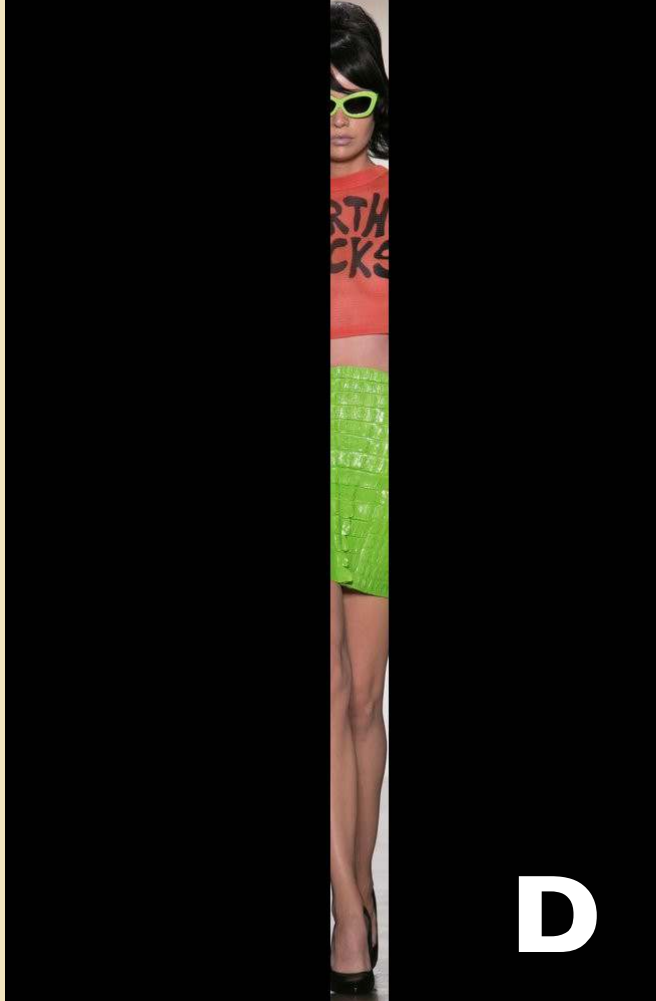
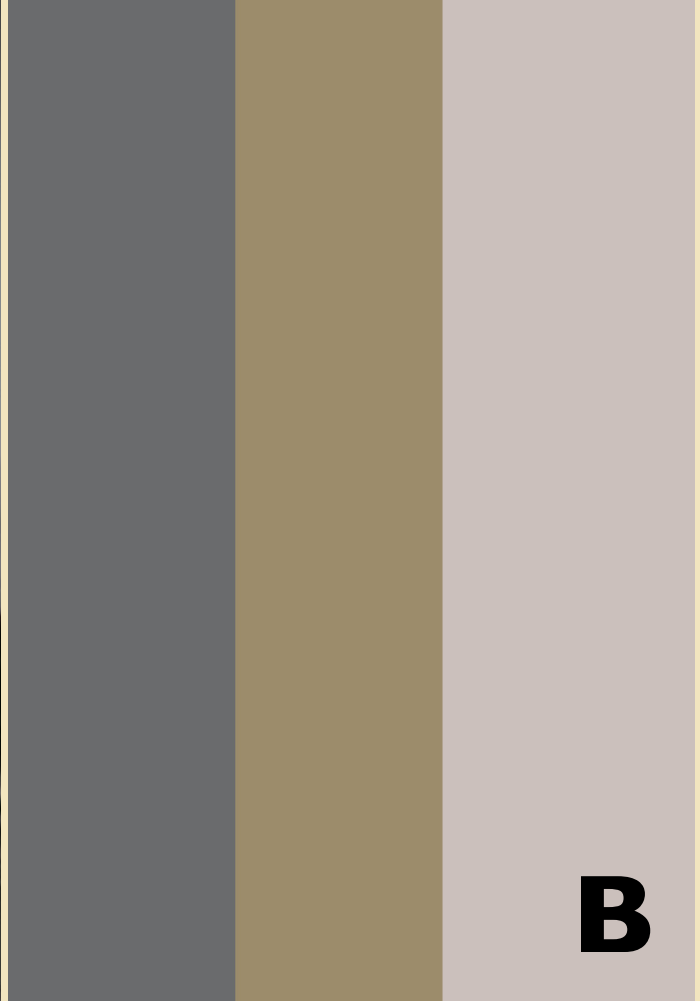
Duckie Brown



Sally La Pointe



Cusnie et Ochs



The pieces

And months of soul searching

- Repa made the processing of 10,000 images at 640x480 easy, fast
- Repa solved resizing issue (do you crop first?)
- Running w/ DevIL
- Color distance and sliding scales of withiness

```
sqcie94 :: (Double,Double,Double) -> (Double,Double,Double) -> Double
sqcie94 (l,a,b) (l',a',b') = lterm ** 2 + cterm ** 2 + hterm ** 2
    where lterm = deltal/(kl * sl)
          cterm = deltac/(kc * sc)
          hterm = deltah/(kh*sh)
          deltal = l - l'
          deltac = c - c'
          c = sqrt (a**2 + b**2)
          c' = sqrt (a'**2 + b'**2)
          deltah = sqrt (deltaa ** 2 + deltab ** 2 - deltac ** 2)
          deltaa = a - a'
          deltab = b - b'
          sl = 1
          sc = 1 + 0.048 * c
          sh = 1 + 0.014 * c
          kl = 2
          kc = 1
          kh = 1
```

$$\Delta E_{94}^* = \sqrt{\left(\frac{\Delta L^*}{k_L S_L}\right)^2 + \left(\frac{\Delta C_{ab}^*}{k_C S_C}\right)^2 + \left(\frac{\Delta H_{ab}^*}{k_H S_H}\right)^2}$$

where:

$$\Delta L^* = L_1^* - L_2^*$$

$$C_1^* = \sqrt{a_1^{*2} + b_1^{*2}}$$

$$C_2^* = \sqrt{a_2^{*2} + b_2^{*2}}$$

$$\Delta C_{ab}^* = C_1^* - C_2^*$$

$$\Delta H_{ab}^* = \sqrt{\Delta E_{ab}^{*2} - \Delta L^{*2} - \Delta C_{ab}^{*2}} = \sqrt{\Delta a^{*2} + \Delta b^{*2} - \Delta C_{ab}^{*2}}$$

$$\Delta a^* = a_1^* - a_2^*$$

$$\Delta b^* = b_1^* - b_2^*$$

$$S_L = 1$$

$$S_C = 1 + K_1 C_1^*$$

$$S_H = 1 + K_2 C_1^*$$

Final Product

Fashion Fingerprints

Color is an important part of the ideas and trends that emerge during Fashion Week. Each look is broken into a minimal stack of color bars, then combined to abstract the collection's hues.



The Book Review

S h h h h

- Awful presentation of back catalog
- Common phrases - failure
- Quotation extraction - interesting
- 40k reviews in ~ 10 sec.
- Exploratory power



Future projects

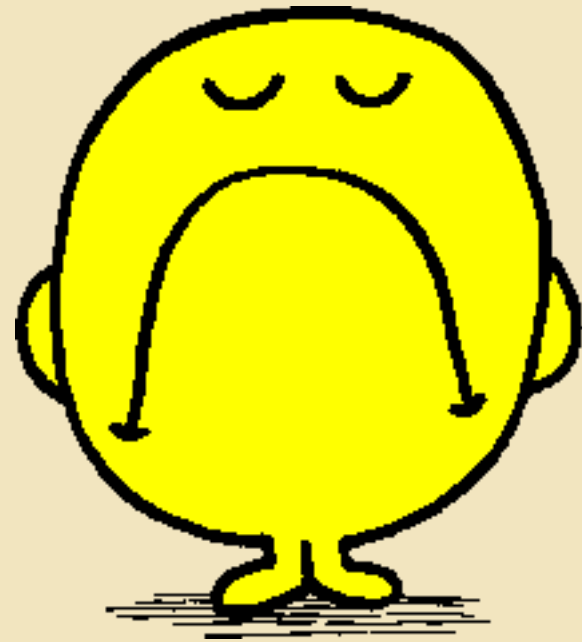
- Replacing our sad, overworked bakers
- Photostream and the agony of a heavy trigger finger (Usain Bolt)
- Our inter app messaging service (probably a pipe dream)



The Dark Side

Not all roses with Haskell

- The tarpits of laziness and profiling as a necessity
- News devs love generic, arbitrary JSON
- Having to hand someone Awodey before I can talk about the lens library type signatures
- Haskell needs that one headline app or framework to motivate the learning curve slog



Op-Eds

More Haskell temptations

- Hackage/ most Haskell documentation blows Ruby's out of the water: types and less folder hell
- The profile grapher
- Type holes and GHCi



Totem and Taboo

H a s k e l l i n a s t r a n g e l a n d

- “What a joke!”
- Hit-by-a-car anxiety
- The sedative of “offline”
- Monadic concerns
- Baby names

QUESTIONS?

- @erikhinton

